Does urbanization filter vegetation through the soil? A spatially validated test with global soil maps in a West African floodplain city

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Abstract

Global soil maps such as SoilGrids 2.0 are increasingly combined with nighttime lights and vegetation indices to infer how urbanization alters soils, and through them vegetation, in data-poor cities, yet this inference is rarely tested. Along the urban gradient of Makurdi, a Guinea-savanna floodplain city in Nigeria, we tested three competing hypotheses — a soil-degradation filter, a geomorphic template and urban amplification of soil sensitivity. We used 3,507 cells of 500 m with VIIRS nighttime lights (NTL), Sentinel-2 EVI and eight SoilGrids properties, and re-examined every inference under geographic validation and spatially robust uncertainty. The SoilGrids built-up mask removed 59% of the most luminous cells, truncating the gradient before the urban core. NTL explained at most 11.5% of the variance of any soil property and about 1% of unique cross-validated EVI variance. The soil-mediated effect of NTL on EVI was -0.010 (block-bootstrap 95% CI -0.057 to 0.035), against a direct effect of -0.088, which rejects the soil-degradation filter. Soil organic carbon, the leading predictor, was negatively related to EVI, indexing hydromorphic floodplain soils. An apparent 2.2-fold steepening of the EVI–pH slope in luminous cells ($p < 0.001$ in conventional tests) vanished within latitude bands ($p = 0.528$), whereas a pH breakpoint at 5.83 (block CI 5.78 to 5.91) was stable. Effective sample size was about 601, and skill fell from $R^2 = 0.914$ in-sample to 0.355 (random) and about 0.2 (geographic cross-validation). We distil these results into a five-step screening protocol for urban uses of global soil maps.

Keywords: SoilGrids, Nighttime lights, Spatial cross-validation, Mediation analysis, Floodplain geomorphology, Guinea savanna

1 Introduction

Urbanization alters vegetation in two ways: directly, by replacing vegetated land with impervious surfaces, and indirectly, by changing the environment in which the remaining vegetation grows (Zhao et al., 2016; Zhang et al., 2022). Global syntheses built on this direct–indirect framework show that the indirect component is often positive, through warming, CO_2 enrichment and management. However, they resolve climatic and management pathways much better than edaphic ones (Zhang et al., 2022; Kang et al., 2025). In the rapidly expanding cities of sub-Saharan Africa, field studies report compaction, nutrient loss and altered soil reaction under urban land use (Asongwe et al., 2022; Rezapour et al., 2025; Zandybay et al., 2024; Fang et al., 2025). Whether such soil change is the route by which urbanization reduces vegetation vigour at landscape scale — an "urbanization filter" acting through the soil (Borowy and Swan, 2022; Kisvarga et al., 2025) — has rarely been tested rather than assumed.

Field soil data are scarce in most African cities, so this question is increasingly addressed with globally or continentally modelled soil products, principally SoilGrids 2.0 (Poggio et al., 2021) and, for Africa, AfSoilGrids250m and iSDAsoil (Hengl et al., 2015, 2021). These are combined with nighttime lights (NTL), vegetation indices and machine learning (Wadoux et al., 2020; Dash et al., 2025; Huang et al., 2025). Urban-gradient studies of this kind usually rank the soil properties that "drive" vegetation along the gradient (Mollashahi et al., 2023; Adelana et al., 2023; Doğan et al., 2025). Yet evaluations of global soil maps in Africa have concentrated on farmland and on-farm decisions (Maynard et al., 2023). Global machine-learning maps are also known to extrapolate silently where training data are sparse (Meyer and Pebesma, 2021, 2022). Whether these products can support inference about urban soil–vegetation coupling has not been examined systematically.

At least five properties of the data and of the analysis can make such inference unreliable:

1. **Coverage bias.** SoilGrids restricts its predictions to land outside built-up, water and glacier classes (Poggio et al., 2021), so the most urbanized cells can drop out of an analysis without notice.

2. **Circularity.** The candidate covariates of SoilGrids include MODIS EVI, NDVI, net primary production and land-cover layers, so an association between modelled soil and observed vegetation is not fully independent evidence.

3. **Confounding.** Cities do not expand at random with respect to landform and drainage, so urban and geomorphic gradients are confounded.

4. **Validation-dependent skill.** The apparent skill of machine-learning models on spatially structured data depends strongly on the validation design (Roberts et al., 2017; Meyer et al., 2019; Ploton et al., 2020).

5. **Spatial dependence in inference.** Thousands of autocorrelated grid cells contain far less independent information than their number suggests, which inflates the apparent significance of interactions and thresholds (Dormann et al., 2007).

This study builds on, and is distinct from, our two earlier papers on the Nigerian Middle Belt. Shaibu and Kirichuk (2026a) evaluated SoilGrids at ward level, in 24 wards of Makurdi and Otukpo, against field vegetation plots. They found weak zonal differences in modelled soil properties except pH, and missing values in the most urbanized wards. Shaibu and Kirichuk (2026b) described multi-scale vegetation responses to urban growth in Makurdi. Neither paper tested whether urban effects on vegetation pass through the soil, quantified masking continuously along the urban gradient, or asked how validation design and spatial dependence change the conclusions. Here we move from wards to 3,507 cells of 500 m. We test the soil-mediated pathway formally against competing hypotheses, and we re-examine every inferential result under geographic validation and spatially robust uncertainty.

We treat the urbanization filter as three competing, testable hypotheses:

- **H1 (soil-degradation filter):** urban intensity shifts soil properties, and these shifts mediate the decline of canopy vigour, so the indirect effect of NTL on EVI through soil is substantial.

- **H2 (geomorphic template):** soil–vegetation relationships are set mainly by pre-urban landform and drainage, so NTL explains little soil variance and the soil-mediated effect is negligible.

- **H3 (sensitivity amplification):** urbanization changes the strength of the vegetation response to soil rather than the mean soil state, producing steeper soil–EVI slopes in more urbanized cells.

The study makes four contributions:

(i) a continuous quantification of the coverage bias introduced by the SoilGrids built-up mask along an urban gradient;

(ii) a variance-partitioning and multiple-mediator test that rejects the soil-mediated pathway, with uncertainty that accounts for spatial dependence;

(iii) a demonstration that an apparently strong urban amplification of vegetation sensitivity to soil pH — highly significant under conventional tests — does not survive geographic controls, whereas a pH breakpoint does;

(iv) a hierarchy of apparent, random, feature-space and geographic validation, together with an estimate of the effective sample size, distilled into a five-step screening protocol for using global soil maps in urban environmental monitoring.

2 Materials and methods

2.1 Study area

Makurdi, the capital of Benue State, Nigeria, lies in the Benue River trough between about 7°34′ and 7°53′ N and 8°22′ and 8°43′ E (Fig. 1). It is a primary commercial centre of Nigeria's Middle Belt, and its built-up area is expanding into the surrounding Southern Guinea savanna (Modu and Yager, 2024; Terkimbi et al., 2025). The climate is tropical wet and dry (Köppen Aw), with a wet season from April to October, mean annual rainfall of 1,200–1,500 mm and mean maximum temperatures near 35 °C in March (Balogun et al., 2019).

The vegetation is a mosaic of gallery forest along the Benue and open parkland savanna dominated by *Parkia biglobosa*, *Vitellaria paradoxa* and *Daniellia oliveri* (Buba, 2015). Ruderal and exotic species such as *Azadirachta indica* and *Eucalyptus* spp. are common in the urban fabric (Ranđelović and Jovanović, 2023). Soils are ferruginous tropical Luvisols and Ferralsols on Cretaceous sandstones, with hydromorphic alluvial soils on the Benue floodplain (Agada et al., 2024; Malum et al., 2025). The river divides the city, and the floodplain extends into the urban core. Urban and geomorphic gradients are therefore spatially entangled by construction, which is the condition under which H1 and H2 are hardest to separate.

2.2 Conceptual framework

Figure 2 contrasts the three hypotheses. Under H1, urban intensification (the anthropogenic driver) alters the soil (the abiotic filter), which in turn reduces canopy vigour (the biological response). Under H2, soil state is set chiefly by landform and drainage, and urban intensity affects vegetation mainly through direct land-cover change. Under H3, urban intensity acts on the link between soil and vegetation. The workflow had six stages (Fig. 3):

1. fusion of NTL, EVI and eight SoilGrids layers in Google Earth Engine (GEE);

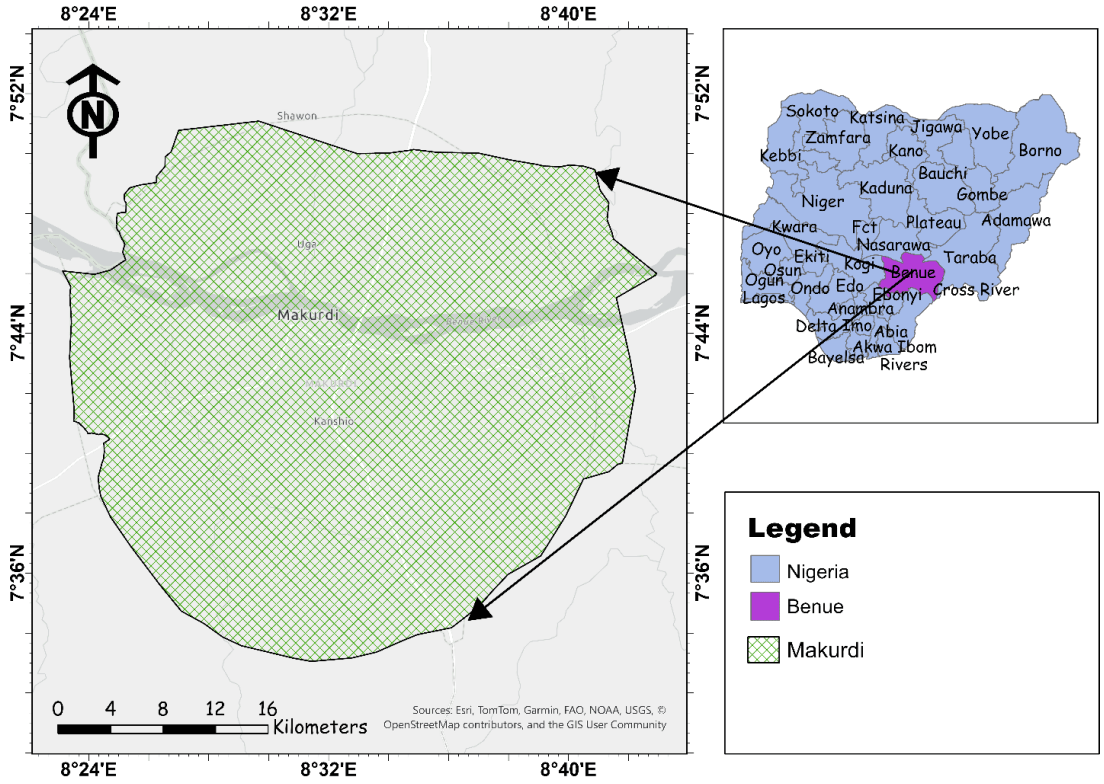


Figure 1: Location of the study area: Makurdi within Benue State and Nigeria, with the delineated study boundary.

2. construction of a 500 m grid and an audit of cells lacking soil values;

3. Random Forest (RF) modelling under a hierarchy of validation designs;

4. tests of H1–H3;

5. breakpoint estimation;

6. re-estimation of every inferential result under spatially robust uncertainty and geographic controls.

2.3 Data acquisition and pre-processing

All layers were processed in GEE (Ghosh et al., 2022).

Urban intensity. We used the VIIRS Day/Night Band monthly cloud-free composites (`NOAA/VIIRS/DNB/MONTHLY_V1/VCMCFG`, band `avg_rad`), taking the per-pixel median of the twelve 2024 composites. Radiance (nW $\mathrm{cm}^{-2}\,\mathrm{sr}^{-1}$) is a continuous proxy for anthropogenic pressure. It is affected by blooming around bright sources and by viewing geometry (Levin et al., 2020; Uyar, 2024).

Soil. Eight SoilGrids 2.0 properties at 250 m were used (Poggio et al., 2021): cation exchange capacity (CEC), soil organic carbon (SOC), total nitrogen (N), pH (H_2O), bulk density (BD), and sand, silt and clay. The 0–5, 5–15 and 15–30 cm layers were averaged (unweighted) and converted to conventional units with the ISRIC factors. Two properties of the product are central here. First, predictions are restricted to land outside the built-up, water and glacier classes of the 2015 ESA land-cover layer. Second,

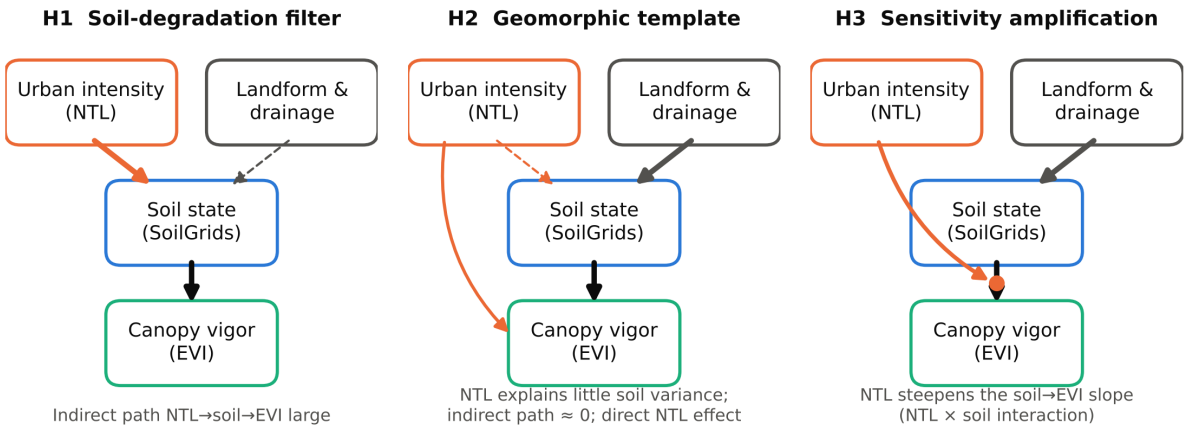


Figure 2: Competing hypotheses. H1: urban intensity degrades the soil, which mediates the decline in canopy vigour. H2: soil state is set by landform and drainage, and urban intensity acts on vegetation directly. H3: urban intensity steepens the soil–vegetation relationship (an interaction). Arrow width indicates the strength of the pathway expected under each hypothesis; dashed arrows indicate weak pathways.

the candidate covariates include land-cover classes, MODIS and Landsat reflectance, and MODIS EVI, NDVI and net primary production (Poggio et al., 2021). SoilGrids layers are therefore spatial predictions with regionally variable uncertainty (Park, 2026), and they are not statistically independent of vegetation indices. Our ward-level evaluation found that they reproduce relative patterns better than absolute values (Shaibu and Kirichuk, 2026a), so all analyses are framed in terms of relative gradients.

Vegetation. The Enhanced Vegetation Index was computed from Sentinel-2 Level-2A surface reflectance as $\mathrm{EVI} = 2.5(\rho_{\mathrm{NIR}} - \rho_{\mathrm{R}})/(\rho_{\mathrm{NIR}} + 6\rho_{\mathrm{R}} - 7.5\rho_{\mathrm{B}} + 1)$, using bands 8, 4 and 2. All 2024 scenes with less than 15% cloudy pixels were averaged. Because such scenes cluster in the dry season, the composite mainly represents dry-season canopy greenness, when edaphic moisture constraints should be most limiting.

Gridding. All layers were summarized on a 500 m covering grid by zonal means. This support reconciles the native resolutions of SoilGrids (250 m) and VIIRS (about 450 m), and no inference is drawn below it.

2.4 Grid, gradient and coverage audit

Discretization produced 3,773 cells. Of these, 266 lacked a value in at least one layer (250 lacked all eight soil layers) and were removed, leaving $n = 3{,}507$. Because the SoilGrids mask excludes built-up and water areas, this deletion is not expected to be random with respect to urbanization. We therefore audited it: we compared the NTL distributions of excluded and retained cells (two-sided Mann–Whitney test) and tabulated the proportion of excluded cells in five NTL classes ($\leq$0.5, 0.5–1, 1–2, 2–4 and $>$4 nW cm^{-2} sr^{-1}).

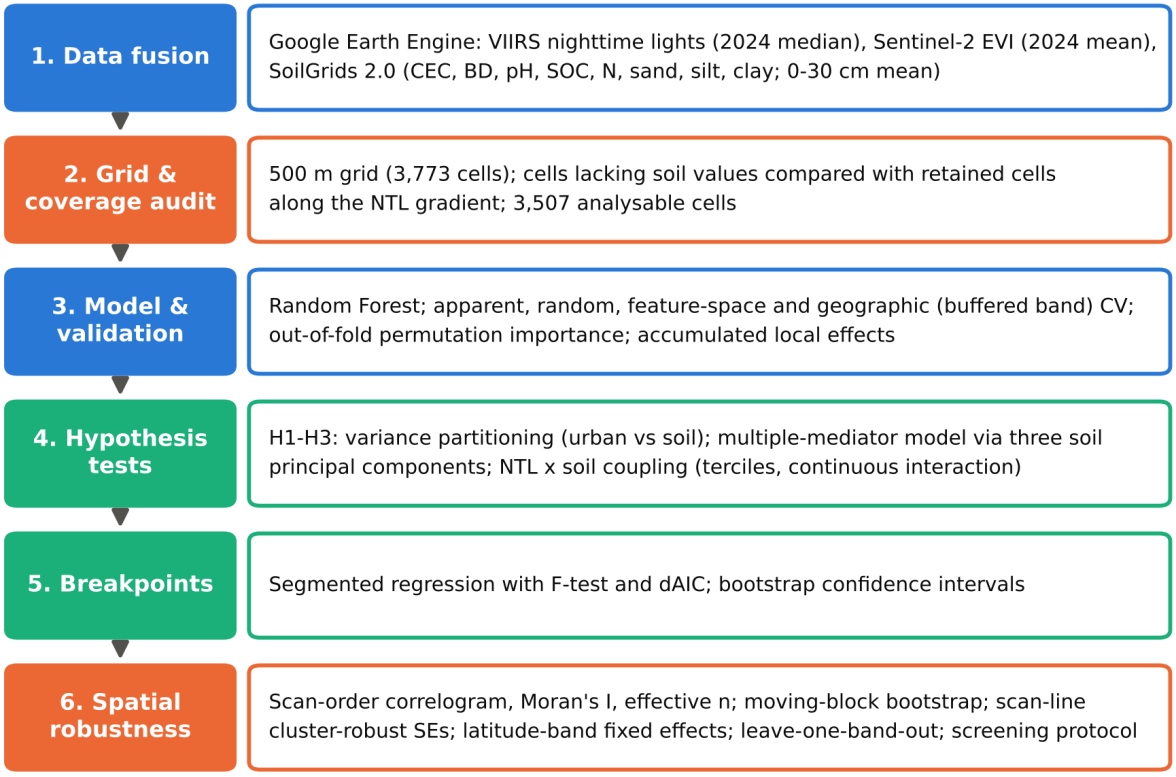


Figure 3: Analytical workflow. Stages 2, 4 and 6 (coverage audit, hypothesis tests and spatial robustness) distinguish this assessment from a conventional driver-ranking analysis.

2.5 Predictive modelling, validation and predictor effects

Pearson correlations described bivariate coupling. Because sand, silt and clay are compositional and the other properties are inter-correlated, the eight standardized soil properties were also summarized by principal component analysis (PCA). The first three components were used in the hypothesis tests.

A Random Forest regressor (scikit-learn; 1,000 trees, unrestricted depth, seed 42) predicted EVI from NTL and the eight soil properties (Simon et al., 2023). Skill was summarized by $R^2 = 1 - \sum_i (y_i - \hat{y}_i)^2 / \sum_i (y_i - \bar{y})^2$ and RMSE $= \sqrt{n^{-1} \sum_i (y_i - \hat{y}_i)^2}$, under five designs:

1. **Apparent fit:** training observations are predicted, to show how far this overstates skill.
2. **Random ten-fold cross-validation (CV):** estimates interpolation within sampled conditions.
3. **Feature-space blocked CV:** withholds ten clusters obtained by k-means on the standardized soil covariates.
4. **Geographic band CV:** withholds ten contiguous latitude bands (Sect. 2.7).
5. **Buffered band CV:** as design 4, but cells within one or two scan lines (about 0.5 or 1 km) of the withheld band are also removed from training (Roberts et al., 2017; Linnenbrink et al., 2024).

All designs were repeated with a regularized forest (minimum leaf size 20).

Predictor importance was estimated by permutation on held-out folds, for single predictors and for groups: NTL; SOC, N and CEC; sand, silt and clay; BD; and pH. Held-out permutation is less biased than impurity importance for correlated predictors (Strobl et al., 2007). Marginal responses were described with accumulated local effects (ALE) (Apley and Zhu, 2020) rather than partial dependence. Partial dependence averages predictions over combinations of correlated soil properties that do not occur in the landscape, and its curves can be unstable (Liang et al., 2023). ALE avoids this by accumulating local prediction differences within quantile bins. Partial dependence is shown only for comparison.

2.6 Hypothesis tests and breakpoints

Variance partitioning (H1 vs H2). EVI variance was partitioned into unique urban, unique soil and shared fractions by comparing models with NTL only, soil only and both (Borcard et al., 1992). Partitioning used cross-validated RF R^2 (minimum leaf size 20; random, feature-space and buffered geographic folds) and adjusted R^2 from ordinary least squares (OLS). We also quantified how much of the variance of each soil property NTL explains.

Multiple-mediator model (H1). Standardized log NTL affected the first three soil principal components (the a paths). These, together with a direct NTL path (c'), affected EVI (the b paths). Specific indirect effects ($a \times b$), their sum and the direct effect received 95% bootstrap intervals (Preacher and Hayes, 2008). Under H1 the total indirect effect should be substantial and have the same sign as the total effect; under H2 it should be near zero.

Gradient-dependent coupling (H3). Cells were divided into NTL terciles: low (0.28–0.39), middle (0.39–0.52) and high (0.52–6.40 $\mathrm{nW\,cm^{-2}\,sr^{-1}}$), with n = 1,171, 1,168 and 1,168. Within each tercile we computed:

- the correlations of EVI with each soil property and principal component;
- the OLS slope of EVI on pH;
- the cross-validated skill of a soil-only RF.

Correlations were compared between the low and high terciles with Fisher's z. The modification of the pH–EVI slope by urban intensity was also tested as a continuous pH $\times$ log NTL interaction, with and without the other soil properties as covariates. Within-tercile standard deviations were compared to exclude range artefacts.

Breakpoints. Single breakpoints in the bivariate EVI response to SOC, pH, BD and log NTL were estimated by segmented regression (Muggeo, 2003). Each breakpoint was located by least-squares grid search over the 5th–95th percentiles of the predictor. Support for a break was assessed with an F-test and with ΔAIC relative to a linear model.

2.7 Spatial dependence and spatially robust inference

The archived analytical table has no coordinates, but its rows are stored in the order in which GEE generates the covering grid, that is, as a raster scan of 500 m cells. We verified this empirically. Consecutive

rows were strongly autocorrelated, and the autocorrelation of NTL peaked again at a lag of 70 rows, the length of one east–west scan line. Cells with EVI < 0.05, which mark the Benue channel, all fell within about 11 consecutive scan lines (half of them within four), consistent with a west–east river crossing a north–south sequence of lines. Contiguous runs of rows therefore form latitude bands of about 54 lines in total. We used them in three ways:

1. as geographic CV folds (ten bands of about 2.7 km, with optional buffers);
2. as clusters for cluster-robust standard errors (Cameron and Miller, 2015), using pairs of scan lines (about 1 km);
3. as fixed effects that remove all between-band (north–south) variation, so that the H3 interaction is estimated only from contrasts within bands.

Residual spatial dependence was summarized with the scan-order correlogram of OLS residuals, along-row Moran's I, and an AR(1) effective sample size, $n_{\text{eff}} = n(1-\rho_1)/(1+\rho_1)$. Bootstrap intervals for the mediation effects, the tercile slope difference and the pH breakpoint were recomputed with a moving-block bootstrap (Künsch, 1989), with blocks of two scan lines, and compared with the independent (i.i.d.) bootstrap. Breakpoint stability was also assessed by re-estimating the breakpoint with each band left out in turn.

Analyses used Python 3 (scikit-learn, statsmodels, SciPy). The scripts that reproduce every number, table and figure are archived as described in the Data and Code Availability statements. Reporting follows the REFORMS recommendations for machine-learning-based science (Kapoor et al., 2024).

3 Results

3.1 Spatial patterns of urban intensity, soil and vegetation

Nighttime radiance formed a concentrated core of high values (>4 nW cm^{-2} sr^{-1}) around the central business district on both banks of the Benue and decayed to below 0.3 at the rural periphery (Fig. 4). EVI broadly mirrored this decline, with the highest values (>0.45) in the peri-urban fringe and riparian gallery forest, but green patches persisted inside the urban core. Modelled soils were organized by the river. Cells near the channel had higher SOC and N, sand dominated the eastern uplands, and clay and silt were concentrated in low-lying depressions. Bulk density showed localized highs (>1.40 g cm^{-3}), and pH was slightly less acidic in and around the luminous core. The maps alone cannot show whether these patterns reflect urban modification of the soil or the position of the city relative to the floodplain; this is tested below. They also show that parts of the brightest core have no soil values.

3.2 Coverage bias of the soil layers

Soil values were missing for 266 of the 3,773 cells (7.1%), and these cells were concentrated in the urban core (Fig. 5). Their median NTL was 2.36 nW cm^{-2} sr^{-1}, against 0.44 in retained cells (Mann–Whitney $p < 0.001$). The proportion of cells without soil values rose monotonically along the gradient:

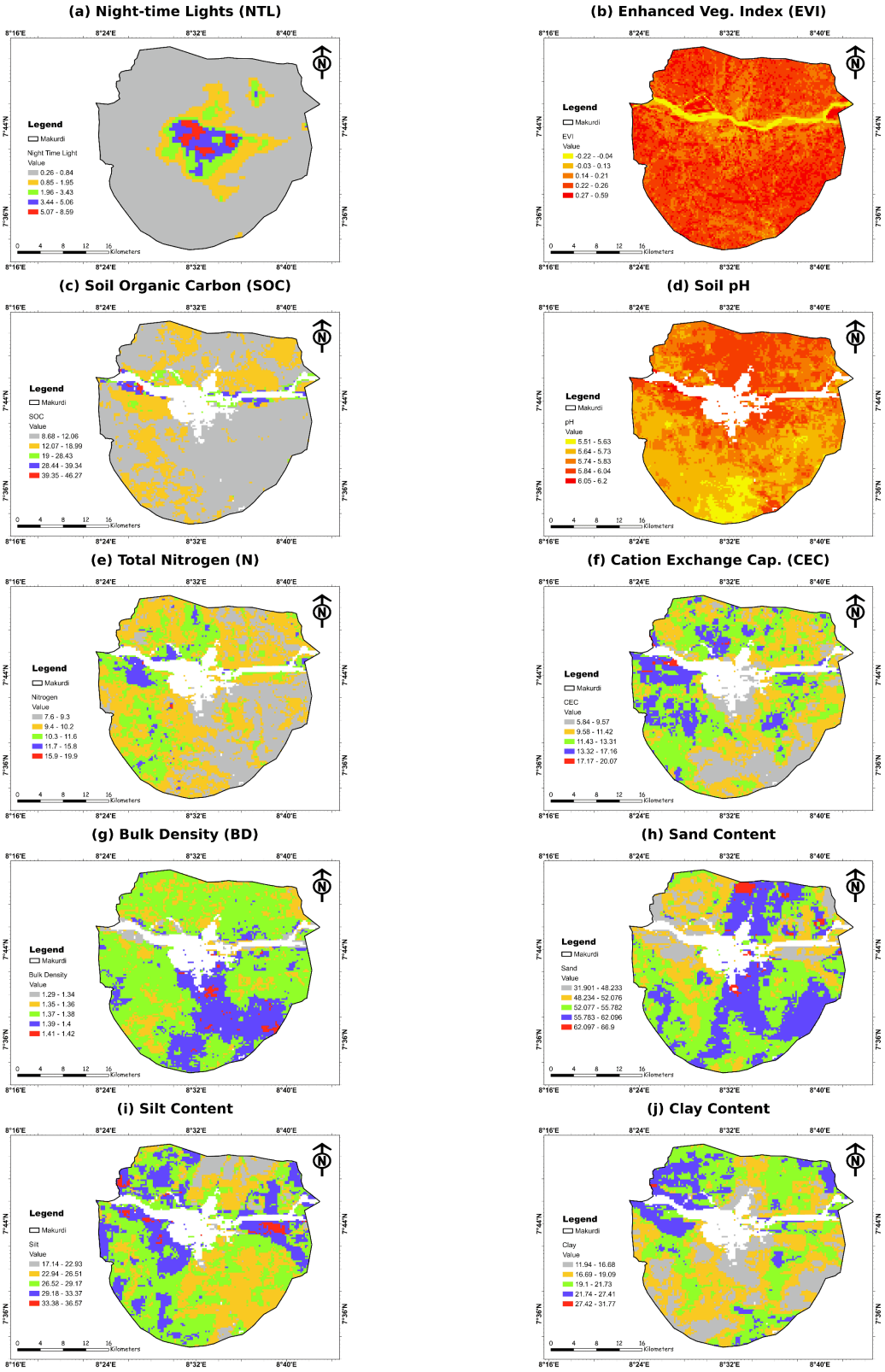


Figure 4: Spatial configuration of urban intensity (NTL), vegetation vigour (EVI) and the eight SoilGrids properties across Makurdi (500 m grid). White areas within the boundary are cells without SoilGrids values (Sect. 3.2).

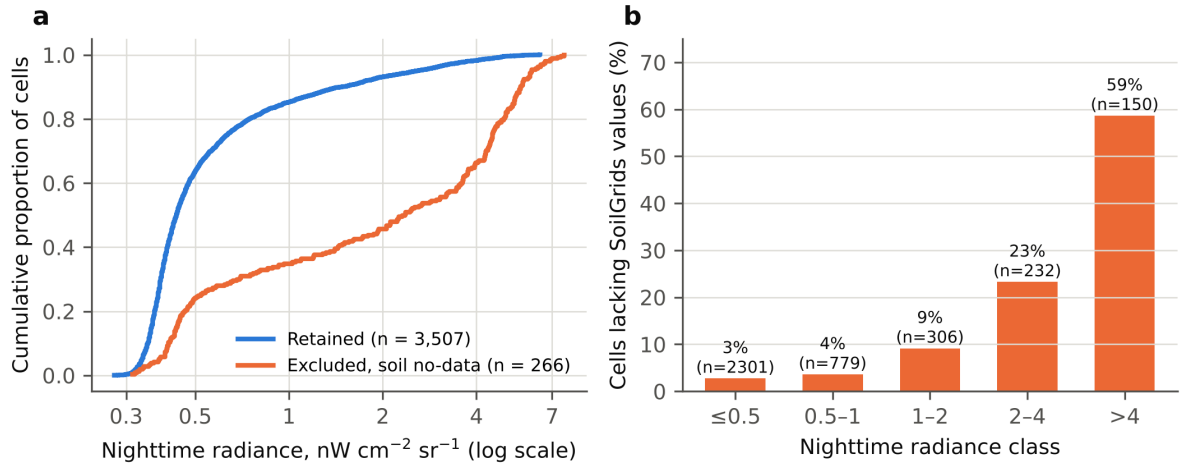


Figure 5: Coverage audit of the SoilGrids layers along the urban gradient. (a) Cumulative distribution of nighttime radiance for retained cells and for cells excluded because SoilGrids values were missing. (b) Proportion of cells lacking SoilGrids values in five radiance classes; *n* is the total number of cells in each class.

NTL class (nW cm^{-2} sr^{-1})	Cells without soil values (%)
≤0.5	3
1–2	9
2–4	23
>4	59

Across the brightest decile (NTL >2.05), 37% of cells were lost. About 104 excluded cells had EVI < 0.05 and correspond mainly to the river channel and bare surfaces; most of the rest lay in the luminous built-up core. The maximum NTL in the analysable sample was 6.40, against 7.64 in the full grid. The analysable gradient is therefore truncated precisely where urban effects on soils would be expected to be largest. All further results describe the peri-urban to moderately urbanized part of the gradient, not the dense core.

3.3 Bivariate coupling

More luminous cells had slightly less acidic (pH: $r = 0.249$), denser (BD: $r = 0.234$) and less clay-rich ($r = -0.339$) modelled soils with lower CEC ($r = -0.253$; all $p < 0.001$; Table 1, Fig. 6). In terms of shared variance these associations are small. NTL explained at most 11.5% of the variance of any soil property (clay), 6.2% of pH and 0.6% of SOC, and an RF using NTL alone explained at most 6.5% of the cross-validated variance of any soil property.

The first three soil principal components explained 45.6%, 17.6% and 11.0% of soil variance:

- **PC1** contrasts coarse, dense, carbon- and nutrient-poor soils with fine, carbon-rich soils, and shared only 6.2% of its variance with NTL.
- **PC2** contrasts silty soils with sandy, less acidic soils.
- **PC3** is dominated by clay.

Table 1: Principal bivariate associations among urban intensity, modelled soil properties and vegetation vigour ($n = 3{,}507$ cells; Pearson r; all nominal $p < 0.001$). Coefficients describe association only.

Pair	r	Interpretation
NTL – pH	0.249	More luminous cells have slightly less acidic modelled soils
NTL – CEC	−0.253	More luminous cells have lower modelled CEC
pH – EVI	−0.349	Less acidic cells have lower vigour (Sect. 3.7)
SOC – EVI	−0.443	Carbon-rich floodplain cells have lower vigour (Sect. 4.2)
NTL – EVI	−0.088	Weak association; soil-mediated path tested in Sect. 3.5

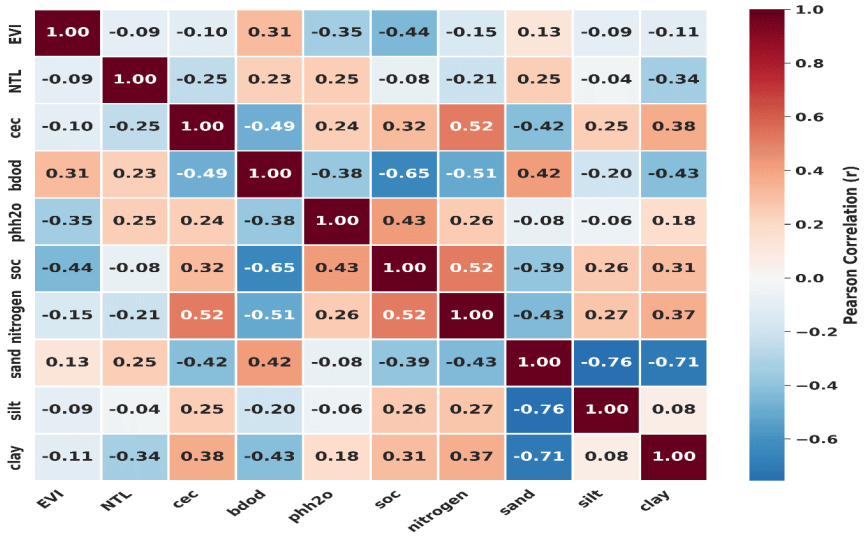


Figure 6: Pearson correlation matrix of urban intensity (NTL), the eight SoilGrids properties and vegetation vigour (EVI) ($n = 3{,}507$).

SOC and pH had the strongest associations with EVI, both negative ($r = -0.443$ and -0.349), whereas BD was positively associated with EVI ($r = 0.314$; Online Resource 1, Fig. S1). Both the SOC–EVI and the BD–EVI signs are opposite to those expected if nutrition and compaction limited vegetation. NTL had only a weak linear association with EVI ($r = -0.088$). With $n = 3{,}507$ even small coefficients are nominally significant; because neighbouring cells are not independent (Sect. 3.8), interpretation rests on effect sizes.

3.4 Predictive skill under a hierarchy of validation designs

Model skill depended strongly on the evaluation design (Table 2; Figs. 7 and 8). The fully grown forest reproduced 91.4% of EVI variance in-sample, a statistic that reflects memorization rather than predictive skill. Random ten-fold CV gave $R^2 = 0.355$. Holding out geographically contiguous latitude bands gave $R^2 = 0.221$, and feature-space blocking gave 0.144. With a buffer of two scan lines around each withheld band, skill was 0.167 for the fully grown forest and 0.194 for the regularized forest.

Geographic skill of the fully grown forest varied with buffer width (0.221, 0.092 and 0.167 for 0, 0.5 and 1 km). The regularized forest was more stable (0.230, 0.227 and 0.194). This indicates that

Table 2: Random Forest performance under different evaluation designs ($n = 3{,}507$). Each design answers a different question; the apparent statistic is not a measure of predictive skill. Geographic designs use 500 trees.

Evaluation design	What it estimates	R^2	
		full depth	leaf $\geq$20
Apparent (in-sample)	Fit to training data, not skill	0.914	–
Random ten-fold CV	Interpolation within sampled conditions	0.355	0.296
Feature-space blocked CV (k-means, $k = 10$)	Transfer to unrepresented soil conditions	0.144	0.203
Latitude-band CV (10 bands)	Transfer to unsampled areas	0.221	0.230
Latitude-band CV, 0.5 km buffer	As above, adjacency removed	0.092	0.227
Latitude-band CV, 1 km buffer	As above, wider buffer	0.167	0.194

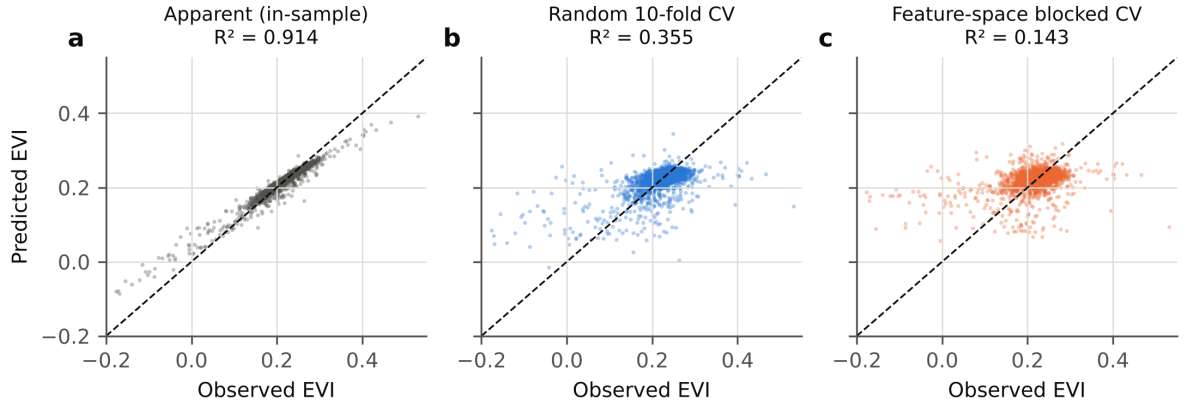


Figure 7: Observed versus predicted EVI under (a) apparent fit, (b) random ten-fold CV and (c) feature-space blocked CV (held-out predictions only in b and c). The dashed line is the 1:1 line.

fully grown trees overfit local structure that does not transfer. In short, about one third of the spatial variance in dry-season EVI can be interpolated from the nine predictors, and about one fifth transfers to geographically separate parts of the landscape.

Out-of-fold permutation identified SOC as the most informative single predictor: the mean drop in held-out R^2 was 0.503 ($\pm$0.147 SD across folds). It was followed by pH (0.156), silt (0.109), total N (0.083), sand (0.073) and NTL (0.054), with CEC, BD and clay contributing least (Fig. 9). Grouped permutation gave the same order: organic matter and nutrients (0.564), texture (0.229), pH (0.156), NTL (0.054) and BD (0.028). Importance measures predictive contribution, not mechanism or direction. SOC is the soil variable that best separates vegetation conditions in this landscape, but its negative sign and its near-zero relationship with NTL ($r = -0.076$) indicate that this separation is not urban in origin.

3.5 Urban versus edaphic control: variance partitioning and mediation

Nearly all explainable variance in EVI was edaphic (Table 3; Fig. 10a). With a regularized RF and random folds, soil alone explained $R^2 = 0.287$ and NTL plus soil 0.297, so the unique urban contribution was 0.010. Under feature-space blocking the values were 0.192, 0.202 and 0.009. Under buffered geographic bands they were 0.221, 0.227 and 0.006. NTL alone had no cross-validated skill under any

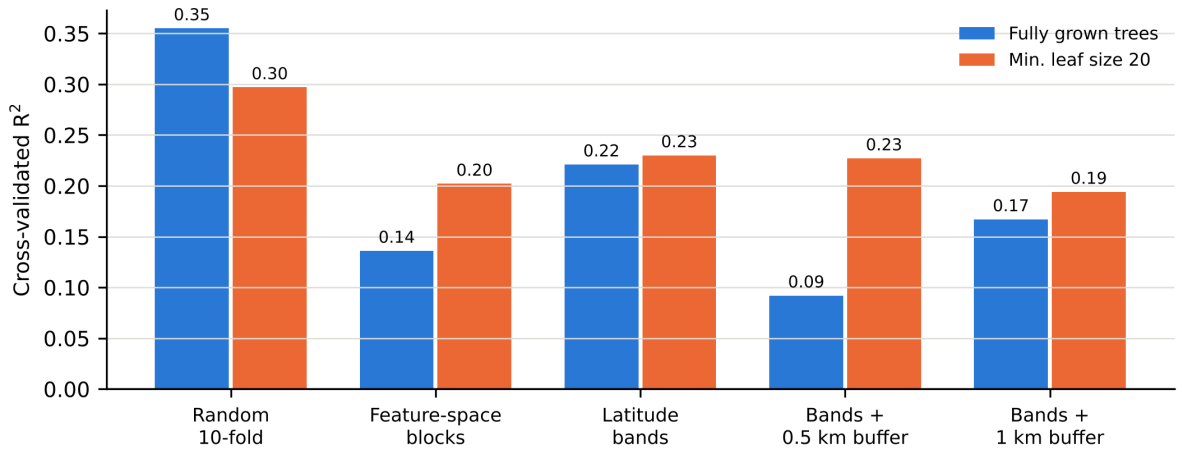


Figure 8: Cross-validated R^2 of the Random Forest (NTL + eight soil properties) under random, feature-space and geographic designs, for fully grown trees and for a regularized forest (minimum leaf size 20). Latitude bands are contiguous runs of the 500 m grid's scan lines; buffers remove the adjacent 0.5 or 1 km from training.

design. The OLS partition agreed: soil alone explained 24.0% (adjusted R^2), NTL alone 0.7% and the full model 24.2%, leaving a shared fraction of 0.5%. Because the shared fraction is the only route by which an urban effect could pass through the soil, H1 predicts it to be large; it was negligible.

In the multiple-mediator model, the standardized total effect of log NTL on EVI was -0.097. Urban intensity was associated with each soil component, with a paths of 0.238 (PC1), -0.183 (PC2) and -0.301 (PC3). Each component was associated with EVI, with b paths of 0.327, 0.222 and 0.156. The resulting indirect effects had opposite signs:

- $+0.078$ through PC1;
- -0.041 through PC2;
- -0.047 through PC3.

They summed to a total soil-mediated effect of -0.010. Its interval included zero under the i.i.d. bootstrap (-0.029 to 0.008) and, more widely, under the moving-block bootstrap (-0.057 to 0.035). The direct effect, -0.088 (block-bootstrap 95% CI -0.141 to -0.025), accounted for about 90% of the total. The data are therefore inconsistent with H1. To the extent that SoilGrids represents soil variation in Makurdi, urban intensity lowers canopy vigour mainly through pathways that do not run through the modelled soil, such as vegetation removal and surface sealing.

3.6 Non-linear responses and breakpoints

Accumulated local effects showed how the fitted RF response changes across each predictor while respecting the correlation among soil properties (Fig. 11). For SOC the response was flat up to about 17 g kg^{-1} and declined above about 20 g kg^{-1}; mean observed EVI was 0.231 in each of the lowest three SOC quintiles and 0.197 in the highest. For pH the response was flat to about 5.8 and declined above it. For BD it rose to about 1.36 g cm^{-3} and then levelled off. For NTL it declined gradually without an

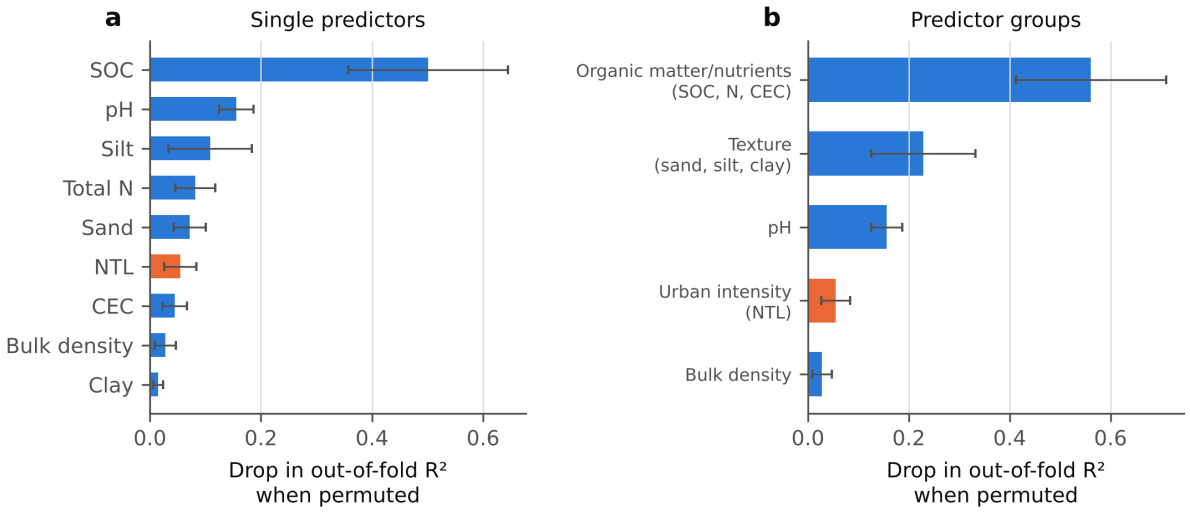


Figure 9: Out-of-fold permutation importance of (a) single predictors and (b) predictor groups: the mean drop in held-out R^2 when the predictor or group is permuted. Error bars are standard deviations across ten folds. NTL is shown in orange.

abrupt change. Partial dependence curves had similar shapes, but their amplitude differed from ALE (for SOC, 0.109 vs 0.096 EVI units). They are therefore not used for thresholds.

Segmented regression supported breakpoints for pH and BD, supported a SOC break only weakly, and found none for NTL (Table 4).

- **pH.** EVI was nearly constant below 5.83 (slope -0.03 per unit) and declined steeply above it (-0.51 per unit; ΔAIC = 292). Mean EVI was 0.231 below and 0.198 above the break.
- **Bulk density.** EVI rose steeply up to 1.360 g cm^{-3} and was nearly flat above (ΔAIC = 287). Cells below this break (13.2% of the landscape) had higher SOC (18.4 vs 11.5 g kg^{-1}) and lower EVI (0.190 vs 0.228).
- **SOC.** The break (13.3 g kg^{-1}) had a wide interval and modest support (ΔAIC = 21).
- **NTL.** A segmented model was no better than a linear one ($p = 0.41$).

The pH breakpoint was insensitive to spatial resampling, with a moving-block bootstrap 95% CI of 5.78 to 5.91. Leaving out each latitude band in turn moved it only within 5.81 to 5.88.

3.7 Gradient-dependent coupling (H3)

Under conventional tests, urban intensity appeared to modify soil–vegetation coupling, most clearly for pH (Table 5; Fig. 12a). The OLS slope of EVI on pH was:

- -0.124 (95% CI -0.149 to -0.099) in the low-NTL tercile;
- -0.171 in the middle tercile;
- -0.277 (-0.306 to -0.248) in the high tercile.

Table 3: Variance partitioning of EVI between urban intensity and soil, and decomposition of the effect of log NTL on EVI into soil-mediated and direct components. RF: Random Forest, minimum leaf size 20. Band CV: latitude bands with a 0.5 km buffer. OLS: ordinary least squares (adjusted R^2). Mediation effects are standardized; "i.i.d." and "block" are 95% intervals from the independent and the moving-block bootstrap.

Variance partitioning (R^2)	RF random	RF feature-blocked	RF band	OLS
NTL + soil	0.297	0.202	0.227	0.242
Soil only	0.287	0.192	0.221	0.240
NTL only	−0.019	−0.057	−0.073	0.007
Unique NTL (full − soil)	0.010	0.009	0.006	0.002
Mediation (log NTL → EVI)	**Estimate**	**i.i.d. 95% CI**	**block 95% CI**	**Share**
Indirect via PC1	+0.078	0.063 to 0.094	–	–
Indirect via PC2	−0.041	−0.051 to −0.031	–	–
Indirect via PC3	−0.047	−0.060 to −0.034	–	–
Total indirect	−0.010	−0.029 to 0.008	−0.057 to 0.035	10% (n.s.)
Direct	−0.088	−0.115 to −0.060	−0.141 to −0.025	90%

Table 4: Breakpoints in the bivariate EVI response estimated by segmented regression, compared with the landscape medians reported as "tipping points" in earlier work. ΔAIC = AIC(linear) − AIC(segmented). The breakpoint CIs are from the i.i.d. bootstrap (500 resamples); the moving-block CI for pH is 5.78 to 5.91. Slopes are in EVI units per unit of the predictor (per log unit for NTL).

Predictor	Median	Breakpoint (95% CI)	Slope below → above	ΔAIC	F-test p
SOC ($g\,kg^{-1}$)	11.27	13.3 (9.6–17.1)	−0.0020 → −0.0056	21.0	< 0.001
pH (H_2O)	5.77	5.83 (5.78–5.88)	−0.034 → −0.508	291.6	< 0.001
BD ($g\,cm^{-3}$)	1.37	1.360 (1.357–1.362)	+2.54 → +0.17	287.1	< 0.001
NTL	0.44	not supported	−0.014 → −0.005	−2.2	0.41

This 2.2-fold steepening was nominally highly significant (tercile interaction and continuous pH × log NTL interaction, both $p < 0.001$). The pH–EVI correlation strengthened from −0.271 to −0.481 (Fisher's $z = -5.94$), and the steepening was not a range artefact: the pH standard deviation was smallest in the high tercile. Urban cells were also more often beyond the pH breakpoint (32.8% of high-tercile cells vs 18.4% of low-tercile cells).

3.8 Robustness to spatial dependence

Residuals were strongly spatially dependent:

- the lag-1 autocorrelation of OLS residuals along the scan was 0.707;
- along-row Moran's I was 0.472;
- the correlogram fell below 0.1 only at about 7.0 km.

Under an AR(1) approximation the 3,507 cells carry the information of about 601 independent observations. Re-estimating the inferential results with this dependence taken into account separated robust from fragile findings (Table 6).

Robust findings. The rejection of H1 held: the total indirect effect remained indistinguishable from zero and the direct effect remained negative under the block bootstrap. The pH breakpoint held (Sect. 3.6).

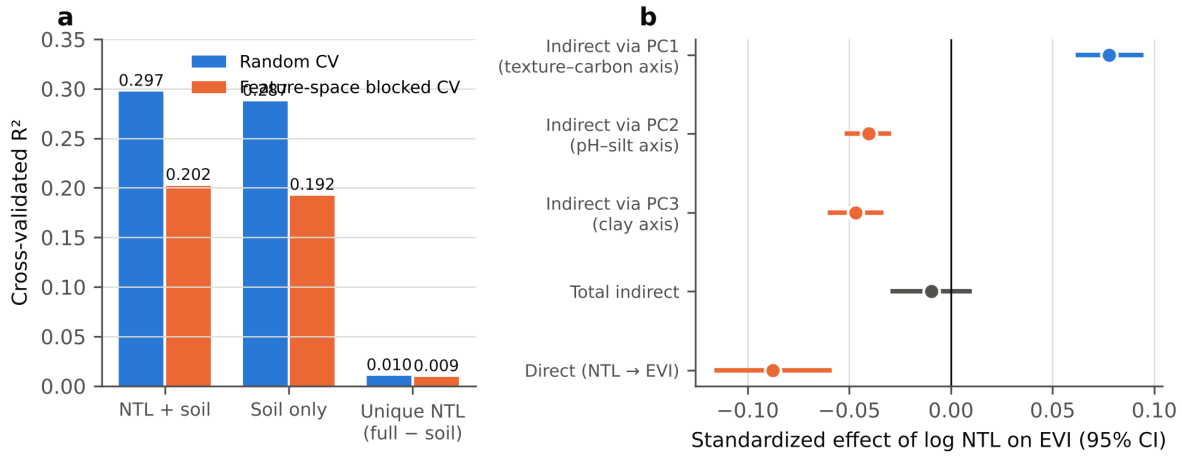


Figure 10: Tests of the soil-mediated pathway (H1). (a) Cross-validated R^2 of regularized RF models with NTL and soil, soil only, and the unique contribution of NTL, under random and feature-space blocked CV. (b) Standardized direct and indirect effects of log NTL on EVI from the multiple-mediator model, with i.i.d. bootstrap 95% intervals (2,000 resamples); grey marks an interval that includes zero. Block-bootstrap intervals are given in Table 3.

Table 5: Soil–vegetation coupling in the low, middle and high terciles of urban intensity. Fisher's z compares the high and low terciles; tests treat cells as independent (see Table 6).

Metric	Low NTL (0.28–0.39)	Mid NTL (0.39–0.52)	High NTL (0.52–6.40)	High vs low
Cells (n)	1,171	1,168	1,168	–
Mean EVI	0.228	0.224	0.218	–
Mean pH (SD)	5.745 (0.089)	5.760 (0.087)	5.802 (0.078)	–
Cells above pH breakpoint (%)	18.4	20.9	32.8	–
EVI–pH slope	−0.124	−0.171	−0.277	nominal $p < 0.001$
r(pH, EVI)	−0.271	−0.292	−0.481	$z = -5.94$
r(BD, EVI)	0.281	0.340	0.384	$z = 2.80$
r(SOC, EVI)	−0.508	−0.407	−0.449	$z = 1.85$
r(PC3, EVI)	0.052	0.137	0.307	$z = 6.39$
Soil-only RF R^2 (CV)	0.258	0.284	0.408	–

Fragile finding. The apparent amplification of pH sensitivity (H3) did not survive. Clustering standard errors on pairs of scan lines inflated the standard error of the pH × log NTL interaction and raised its p value from < 0.001 to = 0.031. Adding latitude-band fixed effects, which removes north–south contrasts between the urban core, the floodplain and the periphery, reduced the interaction to −0.025 (p = 0.528). The block-bootstrap 95% interval for the high-minus-low difference in tercile slopes (−0.153) was −0.264 to 0.032 and included zero. The interaction persisted when the other soil properties were added as covariates (−0.125, $p < 0.001$), so it is not explained by other soil properties; it is largely accounted for by where the luminous cells are.

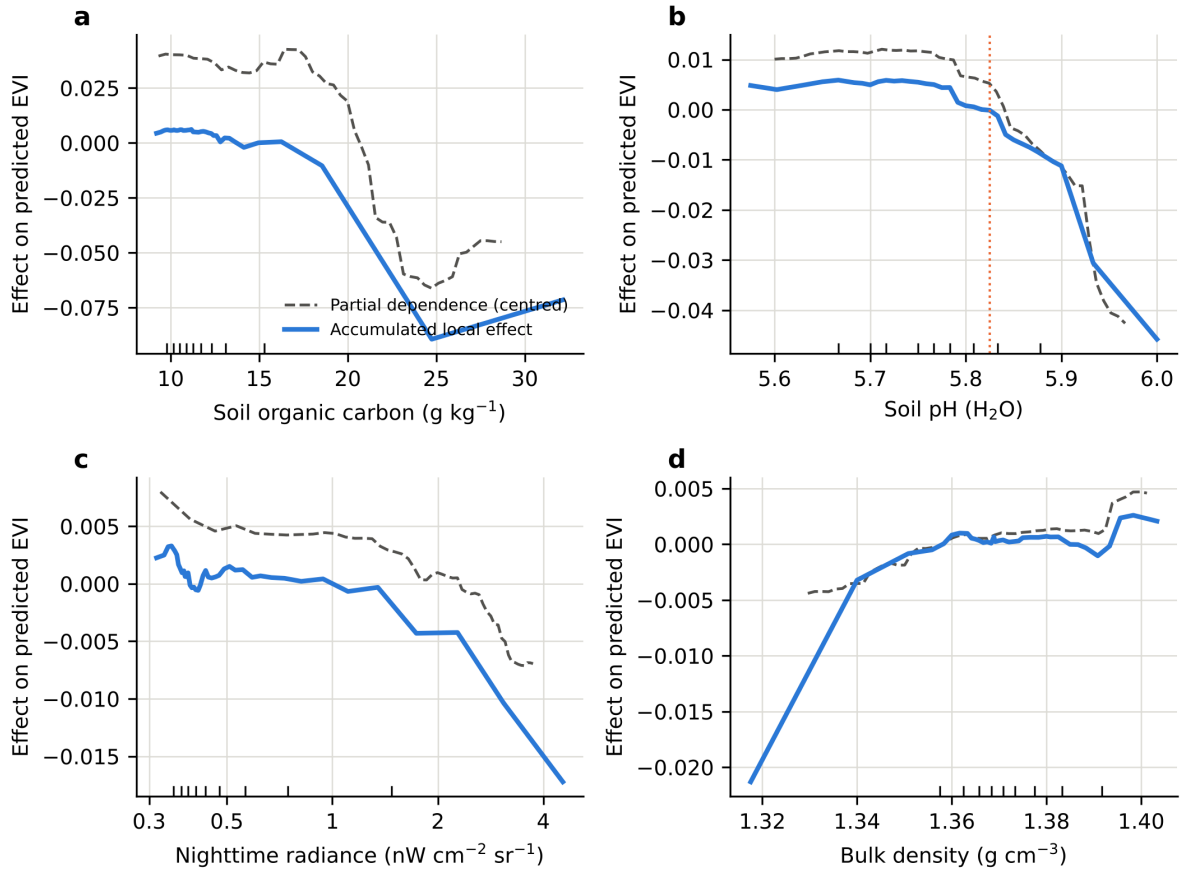


Figure 11: Accumulated local effects (solid) and centred partial dependence (dashed) of predicted EVI on (a) soil organic carbon, (b) soil pH, (c) nighttime radiance (log axis) and (d) bulk density, from a Random Forest with minimum leaf size 5. The dotted line in (b) marks the segmented-regression breakpoint. Ticks on the x-axis mark deciles of the observed distribution.

4 Discussion

4.1 What open soil maps can and cannot see in an African city

SoilGrids does not see the urban core. Its predictions are restricted to land outside the built-up class of a 2015 land-cover map (Poggio et al., 2021), so listwise deletion removed 59% of the brightest cells in Makurdi, and the retained gradient ends before the densest development. Our ward-level study had already noted that the most urbanized wards lacked soil values (Shaibu and Kirichuk, 2026a). The continuous audit shows how steeply the loss rises with radiance and how it truncates the analysable gradient. Any "urban soil signature" recovered from such data describes the peri-urban fringe and scattered built-up land, not sealed or heavily disturbed urban soils. The problem will grow as cities expand beyond the 2015 mask: recently urbanized land keeps its pre-urban soil prediction, so the soil layer lags the city by design.

The second limitation is circularity. SoilGrids models draw on candidate covariates that include MODIS EVI, NDVI and net primary production and several land-cover products (Poggio et al., 2021). The final covariate set differs among properties and is not reported per pixel. An association between modelled soil and observed vegetation is therefore partly an association between vegetation and a func-

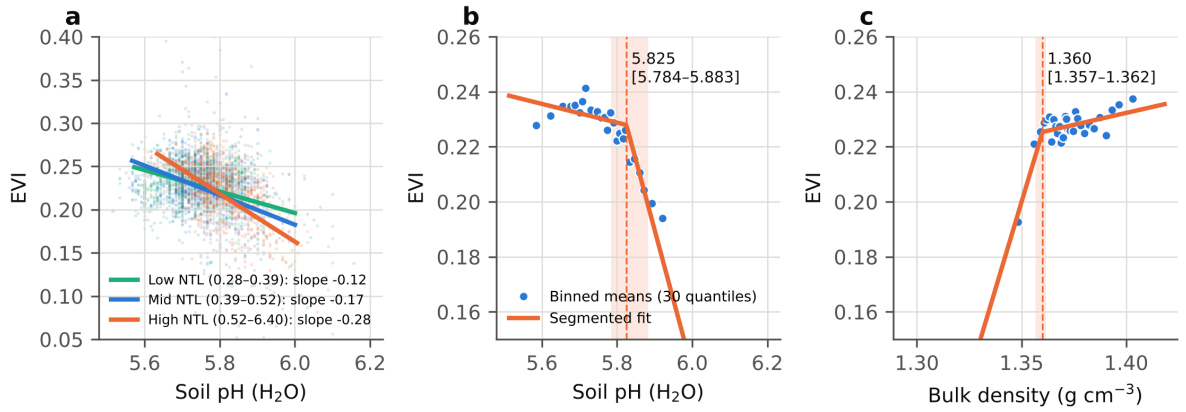


Figure 12: (a) EVI against soil pH in the low, middle and high NTL terciles with OLS fits (slopes in EVI units per pH unit). (b, c) Segmented regression of EVI on (b) soil pH and (c) bulk density; points are means of 30 quantile bins, and the dashed line and shading show the breakpoint and its bootstrap 95% interval. The tercile contrast in (a) does not survive spatial controls (Table 6).

Table 6: Robustness of the inferential results to spatial dependence. Clusters are pairs of scan lines (about 1 km; 27 clusters); band fixed effects (FE) remove variation between ten latitude bands; the block bootstrap resamples blocks of two scan lines. All variables in the interaction models are standardized.

Result	Conventional	Spatially robust	Verdict
Total soil-mediated effect (H1)	−0.010 (−0.029 to 0.008)	−0.010 (−0.057 to 0.035)	H1 rejected
Direct effect of NTL	−0.088	−0.088 (−0.141 to −0.025)	Robust
pH × log NTL (H3)	$p < 0.001$	clustered: $p = 0.031$; band FE: −0.025, $p = 0.528$	Not robust
High−low tercile pH slope	−0.153, $p < 0.001$	block CI −0.264 to 0.032	Not robust
pH breakpoint	5.83 (5.78–5.88)	block CI 5.78 to 5.91; LOBO 5.81 to 5.88	Robust
Unique urban share of EVI variance	0.010 (random CV)	0.006 (band CV)	Robust (small)

tion of vegetation, and should not be treated as independent evidence of edaphic control (Wadoux et al., 2020). Global soil layers are better treated as descriptors of landscape position and broad soil pattern than as measurements of soil condition in individual urban cells.

4.2 A geomorphic template rather than a soil-degradation filter

Within the analysable gradient the evidence favours H2 over H1, and this conclusion was the most robust result of the study:

- urban intensity explained little modelled soil variance;
- it added about 1% of unique explained variance in EVI under every validation design;
- its soil-mediated effect on EVI was indistinguishable from zero under both i.i.d. and spatial-block resampling, because pathways of opposite sign cancelled.

The negative direct effect of NTL is what would be expected if urbanization reduces vegetation mainly through clearance and sealing. This is the direct component of the framework of Zhao et al. (2016) and

Zhang et al. (2022), and it is consistent with direct, non-linear greenness losses reported along other urban gradients (Anwar et al., 2026; Ergen and Gökçe, 2026; Gan et al., 2026). The cascade from urbanization through soil degradation to vegetation decline (e.g. Mollashahi et al., 2023) is therefore not supported by open soil data here. It may still operate at finer scales or in sealed soils that SoilGrids cannot represent.

The negative SOC–EVI and positive BD–EVI associations, both contrary to first principles, are explained by the same geomorphic template. High-SOC cells were clay-rich and sand-poor ($r = 0.309$ and -0.388), of lower bulk density ($r = -0.650$) and higher pH ($r = 0.428$). Cells below the BD breakpoint combined high SOC with low EVI. This is the signature of the poorly drained, fine-textured soils of the Benue floodplain, where seasonal waterlogging, burning and cultivation are likely to suppress dry-season greenness. In this landscape SOC therefore indexes drainage class and topographic position at least as much as nutritional status, consistent with its near-zero coupling with NTL.

Elsewhere SOC responds systematically to land-use intensity (Edmondson et al., 2014; Muzaffar and Saha, 2026; Zhao et al., 2023) and co-varies with canopy condition in Nigerian forests (Mesele and Huising, 2024). In a floodplain city this signal can be overwhelmed by hydromorphy, so the direction of the SOC–vegetation relationship must be established locally rather than assumed. Likewise, within the narrow modelled range (1.30–1.42 $\mathrm{g\,cm^{-3}}$) the positive BD–EVI association reflects the contrast between loose, wet floodplain soils and better-drained uplands. It gives no basis for the compaction limits recommended for temperate cities (Czaja et al., 2020).

4.3 An apparent amplification that did not survive spatial controls

Under conventional tests, vegetation in the most luminous tercile was more than twice as sensitive to modelled soil pH as vegetation in the least luminous tercile ($p < 0.001$). The steepening was not a range artefact and persisted after adjustment for the other soil properties. A plausible mechanism exists: urban soils tend towards higher pH through calcium-rich construction debris (Olchowik et al., 2023), and pH is tightly linked to SOC and nutrient status (Kandadai and Dadhwal, 2026). In an earlier draft of this study we reported this result as the main finding.

It did not survive spatial controls. The interaction weakened when standard errors accounted for spatial clustering. It disappeared when it was estimated only from contrasts within latitude bands, and the block-bootstrap interval for the tercile slope difference included zero. Two features of the landscape explain why. First, the 3,507 cells carry the information of only about 601 independent observations, so nominal p values from cell-level tests are far too small (Dormann et al., 2007). Second, the luminous cells are concentrated in a few bands where the urban core abuts the floodplain. A between-area contrast in the pH–EVI slope, plausibly geomorphic, therefore appears as an urban interaction in a pooled model.

The H3 pattern is thus another expression of the confounding of urban and geomorphic gradients (failure mode 3), revealed only once spatial dependence (failure mode 5) was handled. We report it because the sequence — a highly significant, mechanistically plausible "urban amplification" that disappears under geographic controls — is likely to recur in other urban-gradient studies that treat grid cells as independent.

The pH breakpoint, by contrast, was stable under block resampling and when any single band was left out. It lies close to the landscape median (5.77), which studies of this kind often report as a "tipping point", but the two quantities have different meanings: the median describes where the landscape sits, the breakpoint where the response changes. Because SoilGrids pH varies over only 0.7 units in Makurdi and

Table 7: Five-step screening protocol for using global soil maps in urban-gradient analyses.

Check	What to report	Makurdi outcome
1. Coverage audit	Number and urban-intensity distribution of cells lost to product masks; the part of the gradient to which results apply	59% of cells with NTL >4 lost
2. Circularity	Whether the response (or close correlates) is among the product's covariates; treat soil–response associations as descriptive if so	EVI/NDVI/NPP among SoilGrids covariates
3. Partition and mediation	Unique urban, unique soil and shared fractions; direct test of the soil-mediated path	Indirect effect ≈ 0
4. Validation hierarchy	Apparent, random, feature-space and geographic (buffered) skill side by side	$0.914 \rightarrow 0.355 \rightarrow \approx 0.2$
5. Spatial dependence	Residual autocorrelation, effective n, block-resampled or cluster-robust uncertainty, and within-area (fixed-effects) estimates for interactions and thresholds	$n_{\text{eff}} \approx 601$; H3 not robust

is itself a prediction, the absolute value needs field confirmation. Its relative position — just above the typical modelled pH — is the more robust finding. No breakpoint was supported for NTL, so NTL-based "urban thresholds" should be treated with caution.

4.4 A screening protocol for global soil maps in urban monitoring

The results suggest five checks that can be applied, at little cost, to any urban-gradient analysis built on SoilGrids or similar products (Table 7). Applied in Makurdi, these checks turned an apparently strong urbanization filter — an in-sample R^2 of 0.914, SOC as the dominant "driver" and a highly significant urban amplification of pH sensitivity — into a smaller set of defensible conclusions:

- a geomorphic soil template;
- a direct urban effect on vegetation;
- a stable pH breakpoint;
- honest limits on transferability (about one fifth of EVI variance under geographic validation).

For management in Makurdi, the results support a soil-informed rather than a threshold-driven approach. The floodplain soils identified by the SOC–BD–clay signature are distinctive units whose protection from construction is justified by their hydrological role and carbon storage, not by their canopy greenness (Turrini and Knop, 2015). Planting programmes on less acidic urban soils should include pH testing and avoid incorporating construction debris into planting sites (Malone et al., 2023). Threshold-based planning limits (Liu et al., 2024) should be set only after SoilGrids-based estimates have been calibrated against field survey.

4.5 Limitations and further work

The soil predictors are model outputs, so the analysis characterizes the modelled soil template rather than measured urban soils. Paired field sampling of pH, SOC, bulk density and drainage in urban and rural cells matched for landscape position remains the priority (Maynard et al., 2023).

Geographic blocking here was one-dimensional (latitude bands reconstructed from the grid's scan order). Full two-dimensional blocking, area-of-applicability analysis (Meyer and Pebesma, 2021) and nearest-neighbour distance matching (Linnenbrink et al., 2024) require cell coordinates. So does explicit control for terrain and hydrology — height above nearest drainage, topographic wetness and distance to the Benue — which would test the geomorphic interpretation of SOC directly rather than through soil signatures. A second soil product that does not use vegetation covariates in the same way, such as iSDAsoil (Hengl et al., 2021), would allow a negative-control test of circularity. The archived Earth Engine workflow has been extended to export these layers with coordinates for this purpose.

The mediation model is linear and uses principal components as mediators. The near-zero shared variance in the RF partition suggests that the conclusion does not depend on linearity.

The EVI composite averaged 2024 scenes filtered only by scene-level cloud cover, from a Sentinel-2 collection whose processing baseline changed in 2022. Residual cloud and the reflectance offset bias absolute EVI slightly but are unlikely to alter relative gradients. A pixel-masked, harmonized and seasonally resolved time series would sharpen the dry-season interpretation and allow resilience to be measured directly (Kabano et al., 2021).

Finally, the study covers one city and one year. Applying the protocol across Nigerian savanna cities would show whether the absence of a soil-mediated pathway is general.

5 Conclusions

We tested whether open global soil maps can reveal an urbanization filter on vegetation in Makurdi, a Guinea-savanna floodplain city in Nigeria, and re-examined every inference under geographic validation and spatially robust uncertainty. Five conclusions follow.

1. The SoilGrids built-up mask removed most of the brightest urban cells, so the analysable gradient stops short of the dense core. Coverage audits should be standard in studies of this kind.

2. Urban intensity explained little of the modelled soil variance, and its effect on canopy vigour was direct rather than soil-mediated: the indirect effect was -0.010 (block-bootstrap 95% CI -0.057 to 0.035), against a direct effect of -0.088. The leading soil predictor, SOC, was negatively related to EVI because it indexes hydromorphic floodplain soils, not nutrient supply. The landscape is better described by a geomorphic template than by a soil-degradation filter.

3. An apparently strong urban amplification of vegetation sensitivity to soil pH was an artefact of between-area structure: it disappeared within latitude bands. A pH breakpoint at 5.83 was stable.

4. The 3,507 cells carried the information of about 601 independent observations. Predictive skill fell from $R^2 = 0.914$ in-sample to 0.355 under random and about 0.2 under geographic cross-validation, and no breakpoint in the response to nighttime lights was supported.

5. The combination of global soil maps, nighttime lights and machine learning, now widely used to diagnose urban soil–vegetation relationships in data-poor cities, can produce strong but misleading conclusions unless coverage, circularity, confounding, validation and spatial dependence are examined explicitly. The five-step protocol proposed here is a transferable way to do so.

For Makurdi, the results point to a soil-informed greening strategy that protects floodplain soils, tests pH before planting in the urban fabric, and calibrates all soil-based targets against field survey.

Declarations

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Competing interests The authors have no relevant financial or non-financial interests to disclose.

Ethics approval Not applicable. The study used only publicly available satellite and modelled geospatial data and involved no human participants or animals.

Consent to participate and for publication Not applicable.

Data availability The processed grid-cell dataset, including cells with missing soil values (`data/Makurdi_Full_Suite_EVI_Data.csv`), and a data dictionary are openly available at `https://github.com/shaibu1986/Makurdi_Urban_Ecology`. The release used in this article is archived at Zenodo (**DOI: to be inserted**). Source data are openly licensed products: VIIRS DNB monthly composites (NOAA/EOG), Sentinel-2 Level-2A (Copernicus) and SoilGrids 2.0 (ISRIC, CC-BY 4.0).

Code availability The Google Earth Engine extraction script and the Python scripts that reproduce every statistic, table and figure are in the same repository and archive:

- `scripts/urban_filter_tests.py`: coverage audit, validation designs, importance, partitioning, mediation, coupling and breakpoints;
- `scripts/robustness_scan.py`: geographic validation, spatially robust inference and accumulated local effects;
- `scripts/gee/makurdi_export_v2.js` and `scripts/robustness_geo.py`: the coordinate-based extension described in Sect. 4.5.

Numerical results are written to `results/` and inserted into the manuscript source automatically. Software versions are pinned in `requirements.txt`, and random seeds are fixed.

Author contributions Conceptualization: Ochoche Shaibu, Anatoly A. Kirichuk. Methodology, data acquisition, geospatial processing, formal analysis, software and visualization: Ochoche Shaibu. Validation and resources: Anatoly A. Kirichuk. Writing — original draft: Ochoche Shaibu. Writing — review and editing: Anatoly A. Kirichuk, Ochoche Shaibu. Supervision and project administration: Anatoly A. Kirichuk. Both authors read and approved the final manuscript.

Use of generative AI During the preparation of this work the authors used an AI assistant (Claude, Anthropic) for code review, reproducibility checks, implementation of the spatial-robustness analyses and language editing. The authors reviewed and verified all analyses and text and take full responsibility for the content of the publication.

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